Multi layer cascade failure analysis across urban utility networks

Literature survey:

Paper 1: Predicting Cascade Failures in Interdependent Urban Infrastructure Networks[2025]

- Infrastructures: Power and Road
- Datasets: Synthetic Data (10,227 electric nodes, 4,825 road nodes, and 20,229 communication nodes)
- They use I^3 model. This model combines a method called **Double GAE** to understand multi-layer network maps simultaneously, uses **Multi-Task Pre-training** to ensure the AI understands both local neighborhood dynamics and city-wide structural viewpoints, and applies an **RGCN Decoder** to correctly separate electrical, communication, and traffic relationships as the cascading failure spreads.
- Results: **8.03% to 31.94% boost across classification metrics (AUC, F1-score)** for predicting failure states and a **28.52% reduction in RMSE**.
- Contribution: The core contribution of the paper is pinpointing the exact tipping point where localized infrastructure failures escalate into global, system-wide network collapses.

Paper 2: Vulnerability Analysis of Interdependent Infrastructures Considering the Sensitivity of Components to Different Risks [2023]

- Infrastructures: Power and Water
- Datasets: **IEEE 118-bus power system** which is **synthetic grid blueprint** representing a portion of the 1962 Midwestern US Power Grid and **Anytown and Net3 networks** which are **synthetic/fictional hydraulic benchmarks** originally created by researchers to represent idealized, generic municipal water distribution systems.
- Algorithm: The **Motter-Schneider Load-Redistribution Model** simulates cascading failures by assigning each node a strict capacity limit of $C_i = (1 + \alpha)L_i$, where any component exceeding this threshold via rerouted flow automatically collapses. The algorithm loops through these vertical and horizontal overloads recursively until the entire multi-layer network reaches equilibrium.
- Results:

Paper 3: Vulnerability of Interdependent Urban Infrastructure Networks: Equilibrium after Failure Propagation and Cascading Impacts

- Infrastructures: Transportation, Power and Water
- Datasets**:** The study uses real-world and inferred infrastructure and community data from Maiduguri, Nigeria, Population and community characteristics are incorporated into the model to estimate resource demand and the effects of infrastructure disruptions. The study considers an urban population of approximately 1.2 million
- The authors develop a game-theoretic equilibrium model to represent interactions between infrastructure facilities and communities. Two failure mechanisms are considered: **support failure**, where a facility loses the support it needs from another infrastructure system, and **resource failure**, where a facility cannot meet its resource requirements.
- Following an initial disruption, such as a power-substation failure, the model updates infrastructure availability, resource allocation, congestion, and community choices. A **diagonalization algorithm** based on traffic-assignment methods is used to iteratively calculate the new equilibrium. The process continues until the model converges or reaches its stopping criterion
- Results : A central power substation failure increased water-access costs by 458%, disrupting 62.4% of water infrastructure nodes. Combined water-treatment and fuel-depot failures increased water-access costs by 860%, highlighting the importance of infrastructure interdependencies and resource capacity.

Paper 4: Failure Dependence and Cascading Failures: A Literature Review [2025]

- Objective: Reviews how failures in one component can trigger failures in other interconnected components.
- Methodology: Classifies failure dependencies into direct damage (**Type I**) and load redistribution (**Type II**), reviewing existing cascading-failure models.
- Key Findings: Failure dependencies play a major role in cascade propagation across complex systems.
- Limitations: Highlights the need for better cascade prediction methods and empirical validation.
- Relevance: Provides the theoretical foundation for modelling cascading failures across Bengaluru's road, power and water networks.

Paper 5: Predictive Resilience of Interdependent Water and Transportation Infrastructures: A Sociotechnical Approach [2022] link

- Infrastructures: Transportation and Water
- Datasets**:** Uses real-world infrastructure networks from Tampa, Florida, USA. The study also incorporates U.S. Census data on land use and social vulnerability, along with FEMA flood data from Hurricane Irma (2017).
- The authors construct an interconnected water–road network for Tampa, Florida, incorporating infrastructure topology, land use, and social vulnerability. They use community detection to identify network communities and a Taguchi L_9 experimental design to select combinations of factors for random-failure simulations.
- A Python-based cascading-failure simulator generates resilience outcomes, which are used for predictive modelling. For Hurricane Irma, FEMA flood-depth data and

spatial interpolation identify initially affected infrastructure components, allowing the authors to simulate disaster-driven failures and assess network resilience.

- Results : The study finds that areas with higher social vulnerability are more susceptible to cascading failures caused by both random infrastructure breakdowns and natural disasters. Random failures have a greater impact on residential and institutional land-use areas, whereas natural disasters affect different land-use classes more similarly.
- Contribution: The key contribution is a predictive resilience framework that combines physical, spatial, and social factors to assess cascading failures across interconnected water and road networks. It evaluates both routine infrastructure failures and disaster-related disruptions, helping identify vulnerable areas and support coordinated infrastructure planning.

Paper 6: Resilience Assessment of Urban Connected Infrastructure Networks [2025] link

- Infrastructures: Power, Transportation, Water, Communication and Public Service
- Dataset: The study uses infrastructure-network data for a county in Jiangxi Province, China. The network comprises five infrastructure layers: a power network, a water-supply network, a transportation network, a communication network, and a service network.
- The authors construct a **five-layer network** comprising power, water, transportation, communication, and public services in Jiangxi Province, China. They simulate node failures using random, degree-centrality-based, and betweenness-centrality-based attack strategies.
- As nodes are removed, they calculate the proportion of surviving nodes in the giant connected component (GCC) to measure network reliability. They then compare resilience and recovery performance using **network performance** curves and evaluate how different node-restoration priorities affect the recovery of network connectivity and efficiency.
- Results: The study finds that intentional attacks based on node degree and betweenness centrality can cause greater disruptions than random failures. Networks with multiple interconnected infrastructure layers show lower reliability than individual networks. The study also finds that communication networks exhibit strong resilience and recovery performance, while power infrastructure plays a central role in maintaining connections across the overall system. A betweenness-based recovery strategy restores network efficiency more quickly than a degree-based recovery strategy.
- Contribution: The main contribution is a multilayer resilience assessment framework that combines infrastructure dependencies, cascading-failure simulations, network reliability, and recovery analysis. Unlike studies that focus only on individual infrastructure networks, this work evaluates how disruptions affect an interconnected urban infrastructure system across multiple layers.